AI Team – Intern Engineering – 24 April

1.**OVERVIEW:**

- In this assessment, you are required to build a tool that uses Large Language Models (LLMs) to extract structured information from unstructured resumes (CVs). The goal is to evaluate your ability to work with LLMs, handle PDF files, and design a reproducible evaluation method for your tool.
- You may use any LLM provider (open or closed source), and any orchestration frameworks or tools such as LangChain, LangGraph, CrewAI, OpenAI SDK, etc.
- This is a design-and-build exercise. We are interested in your **end-to-end thinking**: how you process inputs, prompt or design the LLM interaction, validate outputs, and document the solution.
- **NOTE:** Candidates are free tso use AI coding tools (e.g., Claude Code, Antigravity, etc.). We do not evaluate based on whether AI tools are used. ***Instead, we focus on how effectively you leverage these tools and, more importantly, how well you design, explain, and demonstrate that your system works end-to-end thinking.***
- **Time Estimate:** The full test should take **approximately focus times 6-8 hours maybe less since AI Agent Code**. You may take breaks, but please submit your work within **120 hours (5 days)** of receiving the test from Gmail sent.

2. DATASET INPUT:
Use the following resume dataset:
https://www.kaggle.com/datasets/snehaanbhawal/resume-dataset?select=data

Important: Only use files from the **"HUMAN RESOURCES"** and ""**INFORMATION SYSTEMS"** folder. Submissions using files outside this folder will not be accepted.

3. EVALUATION CRITERIA:

- Multiple format handling and input: How well you use the LLM to extract the entity from the unstructured data (plain-text or scanned PDF files)
- Entity Extraction: How accurately you use the LLM to implement the prompt to extract entities and store them in the JSON file
- Product deployments: How can you can serve your ideas as valuable products (UI for users chat, UI for user manage documents, etc.)

4. OUTPUT REQUIREMENTS

I. **RAG Input Handling**
   a. The RAG solution must support both:
      i. Text-based PDFs (digitally generated)
      ii. Scanned PDFs (image-based, requiring OCR)
   b. The RAG solution must support both departments:
      i. Human Resources
      ii. Information Systems

II. **Deployable Product Requirement**
   a. The candidate must demonstrate how the RAG system can be deployed as a user-facing product
   b. Example: Streamlit app or similar lightweight interface or anythings
   c. The product should allow users to upload/query documents and receive responses

III. **Evaluation Requirement** (Optional feature but it's very good to have)
   a. The RAG system must include evaluation methods
   b. Candidates should explain how they measure performance (e.g., accuracy, relevance, latency)
   c. Include both qualitative (manual review) and quantitative metrics (e.g., retrieval precision, answer correctness)

5. Submission:

You need to submit the following items:

- Source Code: You can submit your code in a GitHub repo link
- Documentation (short report or slide): Show your approach to implement this tool (technology stack, workflow, etc.) => ATTACH IT TO README GITHUB